
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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ABSTRACT

Prior work has investigated the computational complexity of deriving two foundational forms of explanations for ML model predictions: (i) *sufficient reasons*, namely subsets of input features whose fixation determines a prediction, and (ii) *contrastive reasons*, namely subsets of input features whose modification alters a prediction. These explanation notions have been studied across several axes, including non-probabilistic versus probabilistic formulations and local versus global regimes. In this paper, we develop a unifying framework in which all of these explanation tasks can be represented as minimization problems over a single probabilistic value function. We then establish that the computational behavior of these tasks is governed by three structural properties of that value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — canonical properties in *combinatorial optimization*. The resulting analysis exposes several counterintuitive structural phenomena. In particular, although the *local* value functions fail to satisfy monotonicity or any submodular/supermodular structure, we prove that the corresponding *global* value functions do satisfy these properties. This separation allows us to derive a sequence of new polynomial-time computability results for globally defined explanations with provable guarantees, across model classes spanning the interpretability spectrum, including neural networks, decision trees, and tree ensembles. Conversely, we show that even highly simplified instantiations of the analogous local explanation tasks remain NP-hard.

1 INTRODUCTION

Despite substantial methodological progress in explaining ML model decisions, a persistent conclusion of the formal explainability literature is that many explanation notions endowed with desirable provable guarantees are computationally difficult to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), with this difficulty typically intensifying for complex or highly non-linear model classes (Barceló et al., 2020; Adolphi et al., 2025; 2024). Consequently, the *computational complexity* of explanation generation has emerged as a central theoretical concern: recent work seeks to delineate, across heterogeneous model families, which explanation guarantees admit efficient computation and which remain computationally inaccessible (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Within the complexity-theoretic study of explanation generation, two particularly fundamental explanation families have received sustained attention: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features S whose fixation to specified values renders the model’s prediction invariant to all assignments of the complementary feature set $\bar{S}$. Conversely, a *contrastive reason* is a subset of input features S whose modification induces a change in the model’s prediction.

In contrast to additive attribution methods, which distribute importance scores over features but can be difficult for users to interpret (Kumar et al., 2020) or insufficiently actionable (Bilodeau et al., 2024), sufficient and contrastive reasons provide *discrete, condition-like explanations* that directly address the questions “what is enough to justify this prediction?” and “what must change to flip

it?”. This conceptual immediacy has made them central to several classical XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018); has been associated with stronger improvements in human prediction relative to additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018); and has supported downstream uses such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A further well-established principle is that *smaller* sufficient and contrastive explanations are generally more interpretable, thereby making *minimality* a central guarantee in the formal analysis of these explanation types (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) conducted one of the earliest studies on the complexity of deriving sufficient and contrastive reasons. Their work established that finding the minimal-sized sufficient reason for a decision tree is NP-hard, with the complexity further increasing to Σ_2^P -hard for neural networks. In the case of contrastive reasons, they demonstrated that computing the smallest possible explanation is solvable in polynomial time for decision trees but becomes NP-Hard for neural networks. Similar hardness results were later shown to hold for tree ensembles as well (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. A common criticism of the classic definition of sufficient and contrastive reasons is their rigidity and lack of flexibility, as they hold in an absolute sense over entire domains and can thus lead to excessively large or uninformative explanations (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To address these limitations, the literature has shifted towards a more general definition that incorporates a *probabilistic* perspective on sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). Under this framework, the goal is to identify subsets of input features that influence a prediction with a probability exceeding a given threshold δ .

Wäldchen et al. (2021) were the first to study the complexity of probabilistic sufficient reasons, showing that for CNF classifiers the problem is NPP^{P} -hard. This hardness extends to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A central theoretical insight in the probabilistic setting is the *lack of monotonicity* in the probability function (Arenas et al., 2022; Izza et al., 2023), which makes even *subset minimal* explanations computationally hard. Strikingly, (Arenas et al., 2022) show that finding a subset minimal probabilistic sufficient reason is NP-hard even for decision trees — unlike in the non-probabilistic case, where such explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. In a more recent study, Bassan et al. (2024) extend the complexity analysis from the *local* (non-probabilistic) setting — where explanations are tied to individual predictions — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons over entire domains. However, as with other non-probabilistic methods (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), the criteria are extremely strict — arguably even more so in the global setting than in the local one. For instance, any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), often making the explanation span nearly all input features and thus less informative.

OUR CONTRIBUTIONS

1. We unify previous explanation computation problems — including sufficient, contrastive, probabilistic, non-probabilistic, as well as local and global — into one framework, described as a minimization task over a unified value function. We then identify three fundamental properties of the value function that significantly impact the complexity of this task: (a) *monotonicity*, (b) *supermodularity*. and (c) *submodularity*.
2. Interestingly, we show that these properties behave in *strikingly different manners* depending on the structure of the value function. In particular, we demonstrate the surprising result that while the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, their *global* counterparts are *monotonic non-decreasing*. Moreover, we identify additional intriguing properties unique to the global setting: the global sufficient value function is *supermodular*, whereas the global contrastive value function is *submodular* — in contrast to the local setting, where neither property holds.

3. We leverage these properties to derive new complexity results for explanation computation, revealing the intriguing finding that global explanations with guarantees can be computed efficiently, even though computing their local counterparts remains computationally hard. We demonstrate these findings across three widely used model types that span the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. First, we prove that while computing a subset-minimal local sufficient/contrastive probabilistic explanation is NP-hard even for decision trees (Arenas et al., 2022), its global counterpart can be computed in polynomial time. We further extend this result to any black-box model (including complex models such as neural networks and tree ensembles) when using empirical distributions. Specifically, we show that obtaining a subset-minimal global sufficient/contrastive explanation is achievable in polynomial time, whereas the local version remains NP-hard for these models.
4. Finally, we present an even stronger complexity result by exploiting the submodular and supermodular properties of the value functions in the global setting — properties that do not hold in the local case. Specifically, we show that it is possible to achieve provable *constant-factor* approximation guarantees for computing cardinally minimal global explanations — even for complex models like neural networks or tree ensembles — when the empirical distribution is fixed. In sharp contrast, we establish strong inapproximability results for the local setting, demonstrating that no *bounded* approximation is possible, even under very simple assumptions.

Because of space constraints, the main text presents the principal theorems and corollaries at a high level, while the complete technical proofs are deferred to the appendix.

2 PRELIMINARIES

Setting. We consider an input space of dimension $n \in \mathbb{N}$, with input vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Each coordinate $\mathbf{x}_i$ may take values from its corresponding domain $\mathcal{X}_i$, which can be either discrete or continuous. The full input feature space is therefore $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$. We consider classification models $f : \mathbb{F} \rightarrow [c]$ where $c \in \mathbb{N}$ is the number of classes. Moreover, we consider a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$ over the input space. In many settings, however, accurately approximating $\mathcal{D}$ is computationally infeasible. A natural alternative is to instead work with a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$, which serves as a practical proxy for the “true” distribution, for instance, by taking $\mathbf{D}$ as a sampled subset from the available training data.

The explanations considered in this work are either local or global. In the local case, an explanation targets a particular prediction $\mathbf{x} \in \mathbb{F}$ and therefore supplies a reason for why the model returned $f(\mathbf{x})$ on that instance. In the global case, by contrast, explanations are intended to characterize the broader behavior of f over a region of the input space, independently of any single $\mathbf{x}$, and thereby to summarize its overall decision-making logic.

Models. Although many of our results are model-agnostic, including the intrinsic structural properties of the value functions, we also derive model-specific complexity consequences for widely used ML model classes. To cover a broad portion of the interpretability spectrum, we analyze models that range from conventionally “black-box” systems to models typically viewed as comparatively “interpretable”. Concretely, we focus on: (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, including majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). Formal definitions are provided in the Appendix.

Distributions. In probabilistic explanation settings, the computational profile may depend substantially on the assumed input distribution $\mathcal{D}$. We therefore distinguish three distributional regimes: (i) *general distributions*, which impose no structural assumptions on $\mathcal{D}$ and hence subsume all possible distributions, and which we use primarily when proving universal properties; (ii) *empirical distributions*, which are induced by a finite dataset $\mathbf{D}$ and constitute a standard practical abstraction in XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, under which the features in $\mathcal{D}$ are mutually independent, another common assumption in the XAI literature (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Importantly, empirical distributions need not encode feature independence; rather, they may preserve complex

dependencies present in finite datasets (Van den Broeck et al., 2022). Complete formal definitions appear in the Appendix.

3 FORMS OF REASONS

This section develops the explanation taxonomy used throughout the paper, beginning with the stringent non-probabilistic formulations of local and global sufficient and contrastive reasons, and then passing to probabilistic variants whose thresholded semantics make the same conceptual guarantees substantially more flexible.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In the setting of *feature selection*, where users frequently retain the top k features deemed most responsible for a model decision, we study the classical *sufficiency* criterion, which underlies several widely used explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022). Such selection may be interpreted locally, with respect to a single prediction, or globally, over the entire input domain. Following standard terminology, we call $S \subseteq \{1, \dots, n\}$ a *local sufficient reason* when fixing the coordinates in S to their values in $\mathbf{x}$ makes the prediction $f(\mathbf{x})$ invariant to every assignment of the complementary features $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff the following condition holds:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the hybrid vector that takes the coordinates in S from $\mathbf{x}$ and the coordinates in $\bar{S}$ from $\mathbf{z}$. Local sufficient reasons are closely related to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek alternative inputs $\mathbf{z}'$ that preserve the prediction, i.e., $f(\mathbf{z}') = f(\mathbf{x})$, while remaining maximally close to the original point. Instantiating $\mathbf{z}'$ as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and measuring proximity by Hamming distance makes the two notions coincide.

A *global sufficient reason* (Bassan et al., 2024) is correspondingly a subset $S \subseteq \{1, \dots, n\}$ whose sufficiency is not tied to a particular instance but instead holds simultaneously for *every* possible input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A second prominent explanatory paradigm identifies feature subsets whose alteration is capable of *changing* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024). The associated objective is to characterize the minimal intervention needed to reverse the model output. Formally, a subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are similarly connected to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment $\mathbf{x}'$ that reverses the prediction, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. Under Hamming-weight distance, this counterfactual formulation coincides with the contrastive-reason definition.

Analogously to the sufficient case, one may ask for a feature subset whose modification changes the prediction for every input in the domain of interest. This global contrastive notion is closely related to methods for detecting bias (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and to *group* counterfactual explanation methods that construct counterfactuals across multiple data instances (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* with respect to f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As discussed in the introduction, non-probabilistic sufficient and contrastive reasons have been criticized for imposing conditions that are often too stringent to be practically informative, thereby motivating *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021). These definitions relax the universal requirements by demanding that the relevant guarantee hold with probability at least $\delta \in [0, 1]$, while the boundary case $\delta = 1$ recovers the original non-probabilistic semantics.

Probabilistic sufficient reasons. We define $S \subseteq \{1, \dots, n\}$ to be a *local δ -sufficient reason* for $\langle f, \mathbf{x} \rangle$ when, after fixing the features in S to their corresponding values in $\mathbf{x}$ and sampling the complementary coordinates from $\mathcal{D}$, the classification remains unchanged with probability at least δ . Equivalently:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ denotes that the features in S of vector $\mathbf{z}$ are fixed to their corresponding values in $\mathbf{x}$. We adopt the standard notion of global explanations — by averaging over all inputs in the global domain — and define a subset $S \subseteq \{1, \dots, n\}$ as a *global δ -sufficient reason* with respect to $\langle f \rangle$ if, when taking the expectation of the local sufficiency probability over samples drawn from the distribution $\mathcal{D}$, the expectation remains with value at least δ . In other words:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. Similar to the non-probabilistic case, we define a *local δ -contrastive reason* as a subset of input features that changes a prediction with some probability. Here, unlike sufficient reasons, we set the features of the *complementary* set $\bar{S}$ to their respective values in $\mathbf{x}$, and when allowing the features in S to vary, we require the prediction to differ from the original prediction $f(\mathbf{x})$ with a probability exceeding δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

For the global setting, and analogously to global sufficient reasons, we call a subset S a *global δ -contrastive reason* by taking the expectation over local contrastive reasons sampled according to $\mathcal{D}$ and requiring this expected value to exceed δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

Across all explanation forms introduced so far—whether sufficient or contrastive, and whether local or global—the literature typically treats explanations of *smaller size* as more interpretable and therefore more desirable, which elevates *minimality* to a central provable guarantee. We consequently analyze two distinct minimality notions that apply uniformly to every explanation type considered here:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the *smallest explanation cardinality* $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal S is subset-minimal, whereas the converse need not hold (see Appendix B.3 for an example). We use the terms subset minimal and cardinally minimal, rather than local and global minima, to avoid terminological interference with local and global explanations, which distinguish instance-level from domain-level reasoning. Both notions apply to *all* explanation forms analyzed here. For example, a cardinally minimal local probabilistic δ sufficient reason is one with minimum possible $|S|$, whereas a subset-minimal one is any qualifying S for which no proper subset $S' \subsetneq S$ also qualifies.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

All of the computational problems introduced above—local or global, sufficient or contrastive, probabilistic or non-probabilistic—can be expressed as the search for a minimal-size subset S satisfying $v(S) \geq \delta$, with the non-probabilistic cases recovered by setting $\delta := 1$. To make this reduction explicit, let v_{uff}^{ℓ} denote the local sufficiency probability from equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and let v_{uff}^g denote its global analogue from equation 6. Likewise, v_{con}^{ℓ} and v_{con}^g denote the local and global contrastive probabilities from equations 7 and 73. This notation yields the following unified formulation of cardinally minimal δ -local/global sufficient/contrastive reasons:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

For the relaxed setting in which the desired explanation is subset-minimal rather than cardinally minimal, the same value-function abstraction induces the following weaker optimization objective:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subseteq S$ it holds that $v(S') < \delta$.

Properties of v that affect the complexity. We next isolate the structural properties of the value function that ultimately govern the complexity of explanation generation. The first property, *non-decreasing monotonicity*, requires every admissible feature addition to have a non-negative marginal contribution $v(S \cup \{i\}) - v(S)$. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The remaining properties, *supermodularity* and *submodularity*, refine monotonicity by characterizing how the marginal contribution $v(S \cup \{i\}) - v(S)$ changes as the conditioning set grows. Supermodularity requires marginal contributions to increase with additional context, whereas the dual submodular condition requires them to decrease. Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

Prior work establishes that, in the local non-probabilistic regime, subset-minimal sufficient or contrastive reasons can be computed by exploiting monotonicity; however, this algorithmically useful structure is available only in the restrictive boundary case $\delta = 1$. From a practical standpoint, that assumption is severe. Although one might expect probabilistic guarantees indexed by arbitrary δ to retain enough structure for efficient computation, existing results show that monotonicity collapses in this setting, thereby making explanation generation computationally harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

At the more extreme end of the spectrum, the *global* non-probabilistic setting studied by Bassan et al. (2024) satisfies a stringent *uniqueness* property: there is exactly *one* subset-minimal explanation. Yet the requirement $\delta = 1$ is especially restrictive globally, since the relevant conditions must hold for *all* inputs. Indeed, Bassan et al. (2024) shows that the unique minimal subset coincides with the set of all features that are not strictly redundant, which may in practice include every feature. We prove that, once the global setting is generalized probabilistically to arbitrary δ , this uniqueness property no longer holds; in fact, the number of subset-minimal explanations can be *exponential*.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Although uniqueness therefore fails outside the non-probabilistic boundary case, we show that the more algorithmically consequential property of *monotonicity* persists for the global value functions at every threshold δ . This holds for both the sufficient value function v_{suff}^g and the contrastive value function v_{con}^g . The result is striking precisely because the analogous local value functions, v_{suff}^ℓ and v_{con}^ℓ , do not satisfy monotonicity:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond the monotonicity separation, we identify additional higher-order structure that is specific to the global setting, namely submodularity and supermodularity. Under the common assumption of feature independence, the global sufficient value function v_{suff}^g is supermodular, whereas the local counterpart v_{suff}^ℓ lacks this property even under the stronger assumption of a uniform, and hence independent, input distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

For the second family of explanation tasks, namely global probabilistic *contrastive* reasons, the corresponding value function exhibits the dual structure: it is not supermodular but rather *submodular*. This conclusion is especially notable because the local contrastive value function satisfies neither submodularity nor supermodularity.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

This section analyzes the complexity of obtaining *subset-minimal* explanations—local or global, sufficient or contrastive—across the model classes considered in the paper. The decisive structural property is the monotonicity of the relevant value function. The previous section established that monotonicity holds for the global value functions v_{con}^g and v_{suff}^g , while failing for the local value functions v_{con}^ℓ and v_{suff}^ℓ . This separation is algorithmically decisive: it explains why a greedy deletion procedure converges to a subset-minimal explanation globally but need not do so locally. We therefore obtain a sharp contrast in which several local subset-minimal explanation tasks are hard, whereas the corresponding global tasks are polynomial-time solvable. We begin with the following generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

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1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$   $\triangleright S$  is a (subset minimal?)  $\delta$ -explanation
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Algorithm 1 is designed to recover a subset-minimal δ -explanation. It initializes S as the full feature set $\{1, \dots, n\}$ and then repeatedly tests whether some feature can be removed while keeping the

value above the threshold δ . At each iteration, the algorithm deletes a feature j whose removal produces the smallest decrease in the value function, equivalently selecting j to maximize $v(S \setminus \{j\})$. When no further deletion is admissible, the remaining set S is returned.

Whether Algorithm 1 actually returns a subset-minimal explanation is determined by the *monotonicity* of the value function v . The algorithm terminates once removing any single feature from S would force $v(S \setminus \{i\})$ below δ ; monotonicity upgrades this one-step certificate into the stronger statement that every proper removal $v(S \setminus S')$ also violates the threshold. Using the monotonicity results established above, we obtain the following proposition:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building on this algorithmic consequence, we next derive model-specific complexity results for several subset-minimal explanation variants within the proposed framework.

Decision trees. We first consider a deliberately simple and ostensibly “interpretable” setting in which f is a decision tree and $\mathcal{D}$ is independent. Even here, a strict separation emerges: Arenas et al. (2022) showed that, unless $PTIME=NP$, computing a subset-minimal local δ -sufficient reason for decision trees under independent distributions is not possible in polynomial time, even under the uniform distribution. By contrast, we show that the corresponding global problem is efficiently solvable; namely, a subset-minimal global δ -sufficient reason for decision trees can be computed in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $PTIME=NP$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. We additionally note that a similar tractability guarantee for computing global explanations also holds for *orthogonal DNFs* (Crama & Hammer, 2011), which we briefly recall are Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. Establishing the result for this class is useful because orthogonal DNFs *generalize* decision trees while preserving their clean structural properties, allowing our guarantees to extend beyond tree-structured models. For completeness, we provide the full argument in Appendix N.

Neural networks, tree ensembles, and other complex models. We then extend the analysis beyond decision trees to substantially more expressive model classes. Under empirical distributions, and assuming only that inference in f is computable in polynomial time, we show that subset-minimal global δ -sufficient and δ -contrastive reasons can still be computed in polynomial time.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

This strong complexity statement, which applies to *any* model under an empirical-distribution assumption, further sharpens the distinction between local and global explanation regimes. For certain complex model classes, subset-minimal *local* explanations remain intractable even under empirical distributions. We prove this for both neural networks and tree ensembles, yielding the following strict separation theorem:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now pass from subset minimality to the more demanding objective of computing *cardinally minimal* δ -sufficient or δ -contrastive reasons. In the global setting, the conjunction of monotonicity with supermodularity or submodularity yields new *provable approximation* guarantees for explanation computation. In the local setting, by contrast, the corresponding tasks remain intractable, thereby further sharpening the separation between global and local explainability.

A unified greedy approximation algorithm. When working with a non-decreasing monotonic submodular function, the problem of identifying a cardinally minimal explanation closely aligns with the *submodular set cover* problem (Wolsey, 1982). In contrast, employing a *supermodular* function leads to a non-submodular variation of this problem (Shi et al., 2021). These problems have garnered significant interest due to their strong approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). In the context of submodular set cover optimization, a standard approach involves using a classic greedy algorithm, which we will first outline (Algorithm 2). This algorithm serves as the foundation for approximating a cardinally minimal sufficient or contrastive δ -reason, and we will later examine its specific guarantees in both cases.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 is structurally analogous to Algorithm 1, except that it proceeds bottom-up rather than top-down. Starting from the empty set, it incrementally augments the candidate explanation by selecting, at each step, the feature that produces the smallest increase in $v(S \cup i)$, and it terminates once every additional feature would push the value beyond δ .

Cardinally minimal contrastive reasons. We begin with global contrastive reasons, where monotonicity and submodularity hold, reducing the task to the classic *submodular set cover* problem and allowing us to apply Wolsey (1982)’s classic guarantee via Algorithm 2. For integer-valued objectives, the algorithm achieves a Harmonic-based factor, and more generally an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ with fixed sample size, this becomes a *constant approximation*, only *logarithmic* in the sample size, yielding a substantially strong guarantee. By contrast, in the *local* setting — even for a *single* sample point — no *bounded* approximation exists, marking a sharp gap between global and local cases.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We also provide a matching *lower bound* for the bound in Theorem 3 in the specific case of an empirical distribution without any independence assumption (see Appendix M).

Cardinally minimal sufficient reasons. Unlike the submodular set cover problem — linked to cardinally minimal global *contrastive* reasons and admitting strong approximations — the supermodular variant, tied to global *sufficient* reasons, is harder to approximate. Still, it offers guarantees when the function has bounded curvature (Shi et al., 2021). The *total curvature* of a function $v : 2^n \rightarrow \mathbb{R}$ is defined as $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Leveraging results from Shi et al. (2021), we show that Algorithm 2 achieves an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

Notably, under a fixed empirical distribution, the approximation becomes constant. While contrastive reasons admit a tighter $\mathcal{O}(\ln(|D|))$ bound, sufficient reasons incur an extra $\frac{1}{1-k^f}$ factor — yet still yield a constant-factor approximation. In sharp contrast, the local variant remains inapproximable, lacking any *bounded* approximation even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In*

contrast, unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

Taken together, these findings strengthen the separation established in Subsection 7.1: not only are subset-minimal global explanations tractable while their local analogues are hard, but cardinally minimal global explanations also admit approximation guarantees whereas the corresponding local variants are inapproximable.

8 LIMITATIONS AND FUTURE WORK

Although many of our central findings — especially those concerning the structural properties of value functions — hold at a general level, we instantiate them only for selected model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions within the proposed framework. Consequently, a number of adjacent settings remain open for systematic analysis. Nevertheless, the results developed here provide foundational insight into the structure of formal explanations, together with new tractability and intractability frontiers that can support broader investigations of explainability scenarios.

More specifically, the study foregrounds several theoretically important open questions:

1. First, it would be valuable to determine whether the tractability results obtained for global explainability can be sharpened for simpler model classes, such as decision trees, linear models, or other architectures with inherently tractable structure. Although local explanations for such models are often computationally intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms studied here exhibit strong structural properties that may permit even tighter complexity guarantees. Refining the approximation guarantees available under empirical distributions is a particularly salient direction.
2. Second, when *exact* expectation computation is infeasible, it remains natural to ask whether alternative approximation techniques — such as Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes like DNF formulas (Meel et al., 2019) (and therefore for tree ensembles), or Monte Carlo–based approximations of the kind used in (Subercaseaux et al., 2025) — can approximate these expectations while preserving meaningful explanation guarantees. The key challenge is to characterize how minimality guarantees degrade under approximation error, and which tolerance regimes still preserve them.
3. Finally, alternative notions of feature importance — for example, loss-based measures such as KL divergence (Conmy et al., 2023) — may exhibit analogous monotonicity, submodularity, or supermodularity phenomena when moving from local to global explanations. Identifying such structure could enable efficient feature-selection procedures under these alternative metrics as well.

9 CONCLUSION

We have presented a unified framework for analyzing diverse explanation notions and, through that framework, exposed a sharp structural contrast between local and global sufficient and contrastive reasons. Although the *local* variants lack the combinatorial-optimization structure provided by monotonicity, submodularity, or supermodularity, their *global* counterparts possess precisely these properties: (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These structural results support a series of complexity-theoretic consequences, showing that global explanations with provable guarantees can be computed efficiently, even for expressive model classes such as neural networks, whereas the corresponding local explanations remain NP-hard even in substantially simplified settings. Overall, the results identify foundational properties of formal explanations and delineate tractable and intractable frontiers for future investigation.

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Appendix

The appendix collects the model and distribution formalizations, auxiliary examples, and complete proofs referenced throughout the paper:

Appendix A provides an expanded discussion of related work on Formal XAI.
Appendix B gives the formal model and distribution definitions used throughout the paper.
Appendix C contains the proof of Proposition 1.
Appendix D contains the proof of Proposition 2.
Appendix E contains the proof of Proposition 3.
Appendix F contains the proof of Proposition 4.
Appendix G contains the proof of Proposition 5.
Appendix H contains the proof of Theorem 1.
Appendix I contains the proof of Proposition 6.
Appendix J contains the proof of Theorem 2.
Appendix K contains the proof of Theorem 3.
Appendix L contains the proof of Theorem 4.
Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work is situated within the broader research program on *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations accompanied by provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central theme in this area is the computational complexity of producing such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because formal explanation generation is frequently intractable for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), a substantial portion of the literature has concentrated on more tractable model classes (Marques-Silva & Ignatiev, 2023), including decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

Beyond complexity-theoretic investigations, substantial effort has also been devoted to computing formal explanations in practice using automated reasoning tools, such as MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) to derive certified explanations for neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Additional approaches seek to alleviate the inherent computational burden through abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, within formal XAI, a related line of work studies methods that leverage combinatorial and submodular optimization techniques to improve the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), particularly for decision tree classifiers. As noted in that literature, as well as in this work, the value-function objective underlying these local explanation variants is not submodular in practice. Nevertheless, greedy algorithms inspired by submodular optimization often serve as effective heuristics and can yield high-quality explanations empirically. In contrast, our work establishes a rigorous theoretical connection between combinatorial and submodular optimization objectives and the *global* variants of sufficient and contrastive reasons. This connection provides formal guarantees and opens the door to a broader range of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

In this section, we formalize the models and distributions referenced throughout the paper. Specifically, Subsection B.1 defines the model families, while Subsection B.2 formalizes the distributions.

B.1 MODEL FORMALIZATIONS

This subsection formalizes the three base-model families analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. We define a decision tree (DT) as a directed acyclic graph that represents a function $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes. The graph encodes the function as follows: (i) Each internal node v is assigned a distinct binary input feature from the set $\{1, \dots, n\}$; (ii) Every internal node v has at most k outgoing edges, each corresponding to a value in $[k]$ assigned to v ; (iii) Along any path α in the decision tree, each variable appears at most once; (iv) Each leaf node is labeled with a class from $[c]$. Thus, assigning values to the inputs $\mathbf{x} \in \mathbb{F}$ uniquely determines a path α from the root to a leaf in the DT, where the function output $f(\mathbf{x})$ corresponds to a class label $i \in [c]$. The size of the DT, denoted $|f|$, is defined as the total number of edges in the graph. To allow for flexible modeling, the ordering of input variables $\{1, \dots, n\}$ may differ across distinct paths α and α' , ensuring that no variable is repeated along a single path.

Neural Networks. We present our *hardness* proofs for neural networks with ReLU activations, though our *tractability results* (i.e., polynomial-time algorithms) apply to any architecture that allows for polynomial-time inference — a standard assumption. Thus, all separation results between tractable and intractable cases we prove carry over to *any neural architecture* at least as expressive as a standard feed-forward ReLU network, encompassing many widely used models such as ResNets, CNNs, Transformers, Diffusion models, and more. Moreover, note that any ReLU network can be represented as a fully connected network by assigning zero weights and biases to missing connections. Following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we thus assume the network is fully connected. Specifically, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP f consists of $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) and one output layer $g^{(t)}$, where each layer is defined recursively as:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$ denotes the weight matrix, $b^{(j)}$ the bias vector, and $\sigma^{(j)}$ the activation function of the j -th layer. Accordingly, the model comprises t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is defined as $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, representing the binary input vector. The dimensions of the network are governed by a sequence of positive integers $d_0, \dots, d_t$, with weight matrices and bias vectors given by $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$, respectively. These parameters are learned during training. Since the model functions as a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers use the ReLU activation function, defined by $\text{ReLU}(x) = \max(0, x)$. Although a sigmoid activation is typically used during training, for interpretability purposes, we assume the output layer applies a threshold-based step function, defined as $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Many popular ensemble methods exist, but since our goal is to provide post-hoc explanations, we focus on the *inference* phase rather than the training process. Our analysis centers on ensemble families that rely on either *majority voting* (e.g., Random Forests) or *weighted voting* (e.g., XGBoost) during inference. However, as with our treatment of neural networks, our *tractability results* — namely, polynomial-time algorithms — extend to any possible ensemble configuration with polynomial-time inference, encompassing an even broader range of ensemble configurations.

Majority Voting Inference. In *majority voting* inference, the final prediction $f(\mathbf{x})$ is assigned to the class $j \in [c]$ that receives the majority of votes among the individual predictions $f_i(\mathbf{x})$ from all $i \in [k]$ (i.e., from each tree in the ensemble).

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ denotes the most frequent label in the multiset S . If there is a tie, it is resolved by a fixed tie-breaking rule (e.g., lexicographic order or predefined priority).

Weighted Voting Inference. In *weighted voting* inference, each model in the ensemble is assigned a weight $\phi_i \in \mathbb{Q}$ representing its relative importance. The predicted class is the one with the highest total weight across all models. Formally, for any $\mathbf{x} \in \mathbb{F}$, we define f as:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the identity function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection gives the distributional formalizations that underwrite the probabilistic explanation definitions and complexity results in the main paper.

Empirical Distributions. The distribution $\mathcal{D}$ over the input features will be defined based on a dataset $\mathbf{D}$ comprising various inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$. Here, the distribution of any given input $\mathbf{x} \in \mathbb{F}$ is defined by the frequency of occurrence of $\mathbf{x}$ within $\mathbf{D}$, specifically by:

$$\mathbf{Pr}(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Formally, given a probability value $p(\mathbf{x}_i) \in [0, 1]$ defined for each individual input feature, we say that the distribution $\mathcal{D}$ is independent if the joint probability over inputs is given by $\mathbf{Pr}(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

We observe that when $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ holds for all $i, j \in [n]$, the distribution reduces to the *uniform distribution*, a special case of independent distributions.

General Distributions. We note that many of the proofs in this work — particularly those concerning fundamental properties of value functions — apply broadly over general distribution assumptions. While empirical distributions (using the training dataset as a proxy) are common in XAI, alternative frameworks for approximating distributions include k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for modeling tabular dependencies (Größer & Okhrin, 2022), or more advanced conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022). Such choices are common in the counterfactual explanation and algorithmic recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which are related to contrastive explanations. Furthermore, when the value function is not computed directly from a structural property of the model (e.g., leaf enumeration in decision trees) or from empirical distributions, one may instead approximate it using methods such as Monte Carlo sampling (Hastings, 1970), following approaches similar to (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

In this subsection, we provide a more detailed discussion of the distinction between cardinal and subset minimality. Cardinal minimality offers a substantially stronger guarantee, as it corresponds to a globally minimal explanation size, whereas subset minimality only ensures a local form of minimality.

To see why cardinal minimality is stronger, note that if $S \subseteq [n]$ is a cardinally minimal explanation, then no smaller set S' with $|S'| < |S|$ can qualify as an explanation. Hence, no strict subset $S' \subsetneq S$ is an explanation, which means S is also subset minimal.

However, the reverse does not hold. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

and the assignment $\mathbf{x} := (1, 1, 1)$, which gives $f(1, 1, 1) = 1$. Fixing feature $\mathbf{x}_1 = 1$ yields a cardinally minimal (and subset minimal) sufficient reason, since the prediction remains 1 regardless of $\mathbf{x}_2, \mathbf{x}_3$. But fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ also gives a subset minimal explanation—yet not a cardinally minimal one. Thus, while every cardinally minimal explanation is subset minimal, not every subset minimal explanation is cardinally minimal.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024). In particular, this phenomenon was shown to occur in functions of the following form. We demonstrate that the same function admits an exponential number of *global* and *probabilistic* sufficient reasons. Notably, the mentioned function is a special case of a broader class of *threshold Boolean functions* described in Wegener (2005). Specifically, for $n = 2k + 1$ with $k \in \mathbb{N}$, this function is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. Notably, the function f is *symmetric*, in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends solely on the number of 1's (or, equivalently, 0's) in the input and is invariant under any permutation of input bits. Furthermore, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function f is symmetric, the condition $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, it can be verified that for any subset $S \subseteq [n]$ of size $k + 1$, we have:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and also set $k := \lfloor \frac{n}{2} \rfloor$. Each of these subsets is *subset-minimal* because any subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ (with respect to the δ threshold). Therefore, we can directly apply the same analysis as in (Bassan et al., 2024). In particular, there are exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. Using Stirling's approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields the corresponding bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. This section establishes monotonicity for both forms of global explanations, sufficient and contrastive. We first prove the property for global sufficiency, beginning with Boolean functions to make the argument transparent, and then generalize it to discrete multi-valued input domains with multiple output classes. We subsequently extend the argument to continuous input domains and show that monotonicity holds for any well-defined classification function. Finally, we transfer the same monotonicity guarantee to global contrastive explanations. Crucially, none of these arguments assumes feature independence: monotonicity holds, in a strong sense, for *any* underlying distribution.

We close the section with a simple example demonstrating that the analogous monotonicity property fails for the local probabilistic sufficient and contrastive value functions.

Lemma 1. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Fix an arbitrary set $S \in [n]$ and a feature $i \notin S$. The following identity holds:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

For notational economy, we occasionally suppress the sampling notation $\sim \mathcal{D}$ in the local probability term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and we write f^+ and f^- for the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We first introduce a technical simplification of the probability term, which will streamline the subsequent monotonicity argument.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the marginal distribution restricted to the coordinates in S . For a point $\mathbf{x} \sim \mathcal{D}$ and a set S , all points $\mathbf{z} \sim \mathcal{D}$ satisfying $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, where $b \in \{0, 1\}$, share the same conditional probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. For every $\mathbf{x} \sim \mathcal{D}$ with the same x_S and the same output b , marginalizing over $x_{\bar{S}}$ gives $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, and therefore:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2],
\end{aligned} \tag{22}$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. We now proceed to prove the monotonicity claim, building on the previous simplification. We begin by introducing a few additional notations. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$. Similarly, define $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\ &\geq 0. \end{aligned} \tag{26}$$

Note that $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and similarly: $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Using these identities along with the previous result, we conclude that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby concluding our proof. $\square$

Lemma 2. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. Extending the proof from the simplified Boolean case, we now generalize the result to functions whose input domains and output ranges each consist of a finite set of values. Let $\mathcal{K}$ represent the set of output classes.

We can generalize equation 22 as follows:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] &= \\ \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] \end{aligned} \tag{27}$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] &= \\ = \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right]. \end{aligned} \tag{28}$$

Let P_k denote $\mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Hence, $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, which completes the proof for this setting. $\square$

Lemma 3. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend the monotonicity results to functions where features are defined over continuous domains. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in lemma 2 by revising the notation accordingly. The core reasoning remains unchanged when transitioning from the discrete to the continuous case—only the notation differs.

Alternatively, we provide a direct proof below. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the corresponding joint probability density function. Moreover, let $p(x_i)$ denote the marginal probability density function of x_i , and $\mathcal{X}_i$ denote the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t \mid \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. As a consequence of lemmas 1 to 3, we conclude that the global sufficient value function v_{suff}^g is monotonic non-decreasing for any well-defined classification function — whether its features are Boolean, discrete, continuous, or a combination — under any data distribution.

$$v_{\text{con}}^g(S) = 1 - v_{\text{suff}}^g(\bar{S}) \quad (38)$$

This implies that v_{con}^g is monotonic non-increasing with respect to adding features to $\bar{S}$. Equivalently, v_{con}^g is monotonic non-decreasing as features are removed from $\bar{S}$ and added to S , thus completing the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This section proves supermodularity for the relevant global explanation value functions. As in the monotonicity analysis, we begin with Boolean functions and then progressively generalize

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

to multi-valued discrete domains, continuous domains, and finally arbitrary well-defined classification functions. Unlike monotonicity, however, these supermodularity arguments require feature independence; we therefore also include a counterexample showing that the property can fail once this assumption is removed.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Particularly, we will prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where $S \subseteq S'$ and $t \notin S'$. That is, $\Delta_t(S, f)$ is monotone.

Clearly, if $S' = S$, then $\Delta_t(S, f) \leq \Delta_t(S', f)$ holds. Let $S' = S \cup \{i\}$, where $i \notin S$ and $i \neq t$. For the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, under the assumption of feature independence, we have that the following holds:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ = \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This implies that $\Delta_t(S, f)$ is monotone, that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote $\Pr(z_i = v_{k'})$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$. We

decompose equation 30 to

$$\begin{aligned}\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \left(\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right].\end{aligned}\quad (44)$$

Likewise, we have

$$\begin{aligned}\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].\end{aligned}\quad (45)$$

Therefore, we get

$$\begin{aligned}\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\ &\geq 0.\end{aligned}\quad (46)$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that $\mathcal{D}_p$ satisfies feature independence. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we provide a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \quad (47)$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \quad (48)$$

We let $Y_j(i, t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
\Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\
&= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}.
\end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned}
\Delta_t(S', f) &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \tag{50} \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S,
\end{aligned}$$

where

$$\begin{aligned}
&\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\
&= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}.
\end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned}
\Delta_t(S', f) - \Delta_t(S, f) &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&\geq 0.
\end{aligned} \tag{51}$$

Consequently, the global sufficient value function v_{suff}^g retains the supermodularity property for any valid classification function, regardless of whether its features are Boolean, discrete, continuous, or a mix thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \tag{52}$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , we have that the following holds:

$$v_{\text{suff}}^l(\emptyset) = \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4},$$

$$v_{\text{suff}}^l(\{1, 2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1.$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. $\square$

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once we establish that the global sufficient setting is supermodular, it directly follows that the global contrastive setting is submodular. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. To compare the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently compare $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, hence concluding this proof.

We will now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$v_{\text{con}}^l(\emptyset) = 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2},$$

$$v_{\text{con}}^l(\{1, 2\}) = \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}.$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. $\square$

In the rest of this section, we show that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Moreover, define a distribution $\mathcal{D}$ on x_1 and x_2 . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, we have that the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1)) \\ &= (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1) \end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned} & \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\ &= \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\ & \quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) \end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0)} + \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 1, x_2 = 0)} + \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 1)} \\
& = 1
\end{aligned}$$

Let $t = 2$. Evidently, we have $\Delta_t(\{1\}, f) = 0$. Next, let us analyse $\Delta_t(\emptyset, f)$. We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2 \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \Pr(x_1 = 0)^2 + \Pr(x_1 = 1)^2.
\end{aligned}$$

This implies $\Delta_t(\emptyset, f) = 0$, that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. However, without the assumption of feature independence, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. For example, let:

$$\Pr(x_1 = 0, x_2 = 0) = P_{00}, \Pr(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\Pr(x_1 = 1, x_2 = 0) = P_{10}, \Pr(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. We note that this proposition follows directly from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (as established in Proposition 2), along with other relevant conclusions drawn in prior work. In the remainder of this section, we elaborate on how this monotonicity property leads to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g follows directly from the monotonicity property, for which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) showed that this type of greedy algorithm yields a subset-minimal explanation when the underlying value function is monotone non-decreasing. The algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ drops below δ , ensuring that although $v(S) \geq \delta$ (a maintained invariant), removing any element causes the condition to fail. Due to monotonicity, this implies that no proper subset of S satisfies the condition, guaranteeing that S is subset-minimal.

We now turn to explain why Algorithm 1 does not converge to a subset-minimal solution in the local setting. We note that it is well known that the local probabilistic sufficient value function (and likewise the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables $\{x_1, x_2\}$ and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$ which is classified as 1, then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , the S denotes the set of features that are allowed to vary.) Because $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is also non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We begin by noting that for decision trees, assuming that $\mathcal{D}$ represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

Can be carried out in polynomial time using the following procedure. Thanks to the tractability of decision trees, we can iterate over all pairs of leaf nodes, each corresponding to partial assignments

$\mathbf{x}_S$ and $\mathbf{z}_{S'}$. For each such pair, if both leaf nodes yield the same prediction under f , we compute the corresponding term in the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, allowing the full expectation to be computed by summing the contributions of all such matching leaf pairs.

Since all probabilities involved are provided as part of the input and each step involves only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step in Algorithm 1 is efficient, and thus the algorithm as a whole runs in polynomial time. Finally, combining this with Lemma 2, which proves that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation. This is because the algorithm halts when, for all $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , while $v(S) \geq \delta$ is preserved throughout. By monotonicity, this guarantees that no strict subset of S satisfies the condition, ensuring S is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming $P \neq NP$, there is no polynomial-time algorithm for computing a subset-minimal δ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

Can be performed in polynomial time when $\mathcal{D}$ is selected from the class of empirical distributions. This is achieved by iterating over pairs of instances $\mathbf{x}, \mathbf{z}$ within the dataset and running an inference through $f(\mathbf{x})$ and $f(\mathbf{z})$ to compute the expected values at each step. Consequently, determining whether the probability function exceeds or falls below a given threshold δ can also be accomplished in polynomial time. Furthermore, leveraging the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we can utilize algorithm 1 for contrastive reason search or for sufficient reason search. By executing a linear number of polynomial queries, a subset-minimal explanation (whether a subset minimal contrastive reason or a subset minimal sufficient reason) can thus be obtained in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $P = NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part of the proof follows directly from Proposition 6, which established that this holds for any model, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ can be computed in polynomial time from

the empirical distribution. For the second part, we proceed by proving two lemmas: one establishing the intractability of computing the sufficient case, and the other addressing the contrastive case.

Lemma 6. *Unless $P_{TIME} = NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local δ sufficient reason for neural networks, then it would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution $\mathcal{D}$ consists of a single element, which, for simplicity, we take to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to $\mathbf{0}_n$, we define $\mathbf{1}_n$ as an n -dimensional vector consisting entirely of ones. Given an input ϕ , we initially assign $\mathbf{1}_n$ to ϕ , effectively setting all variables to True. If ϕ evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

ϕ' , while no longer a CNF, can still be transformed into an MLP f using Lemma 8, ensuring that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Given our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we can determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, noting that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We now assert that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Let us assume that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is $\emptyset$ in this case, this trivially holds. The fact that S is subset-minimal means that any other subset $S' \subseteq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution contains only the point $\mathbf{0}_n$), the probability function can only take values of 1 or 0. Hence, we also know that $v(S') = 0$. This tells us that, aside from the subset $S = \{1, \dots, n\}$, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the rest to 0 does not result in a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly implies that ϕ is assigned False for any of these inputs. Since we already know that ϕ does not return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now, suppose that S is a subset that is strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we can apply the same reasoning as before to

conclude that $v(S) = 1$. This implies that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False. Consequently, the ϕ component must be True, which implies that $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We observe that we can apply the same process used in our proof for neural networks, where we encoded ϕ' into an equivalent neural network. However, ϕ' is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since ϕ' consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. *Unless $\text{PTIME} = \text{NP}$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of ϕ .

First, we check whether assigning all variables in ϕ to 1 evaluates ϕ to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network f . Next, we compute a subset-minimal δ -sufficient reason by setting $\mathcal{D}$ as the empirical distribution containing only a single data point $\mathbf{0}_n$, following a similar procedure to Lemma 6. Additionally, we set $\delta = 1$ and compute a subset-minimal δ -contrastive reason concerning $\langle f, \mathbf{x} \rangle$.

We will now demonstrate that if any subset-minimal δ contrastive reason obtained for ϕ is satisfiable, then ϕ itself is satisfiable. Conversely, if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$, where the distribution $\mathcal{D}$ considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a $\delta = 1$ contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of f , which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates f (and consequently ϕ) to True, implying that ϕ is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We decompose the theorem into two lemmas: one establishing the global approximation guarantee and the other proving the absence of any bounded approximation in the local setting.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We prove a slightly stronger statement: the claim holds for *any* model whose inference can be performed in polynomial time, together with one additional mild condition specified below; both requirements are satisfied by our neural-network and tree-ensemble formalizations.

We begin by observing that, because the distribution is empirical, the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

For the approximation guarantee, the classical result of Wolsey et al. (Wolsey, 1982) gives a harmonic-series approximation bound for integer-valued monotone non-decreasing submodular functions v . In its more general form, the bound is $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. Proposition 2 establishes that v_{con}^g is monotone non-decreasing, and, under feature independence, Proposition 4 establishes submodularity. Together with the polynomial runtime of Algorithm 2, these facts immediately yield the approximation factor $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must also show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|D|))$, implying that it is effectively constant, due to the assumption of a fixed dataset $|D|$. To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to be zero. We begin by formally defining what we mean by redundancy:

Definition 1. *Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.*

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ indicates that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ indicates that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, in this sense, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then for any input of size n' , the output of f' will exactly match the output of f when applied to the same features $[n] \setminus \{i\}$. This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles,

it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as previously discussed in this proof and in Proposition 2 — we can compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in \mathcal{D}$ and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature i is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then i is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution $\mathcal{D}$, we can remove them and construct an equivalent model f' . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Therefore, once the preprocessing step removes all redundant features from f and we construct the resulting function f' , we ensure that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|\mathcal{D}|))$. Given that the empirical distribution $\mathcal{D}$ is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|\mathcal{D}|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|\mathcal{D}|^2}$. In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

Which concludes our proof.

□

Lemma 11. *Unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.*

Proof. Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble—even when $|D| = 1$. However, Lemma 9 establishes that even with a single baseline $\mathbf{z} = 0_n$ (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^T} + \ln \left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We again split the argument into two lemmas, the first proving the global approximation guarantee and the second establishing the corresponding local intractability result.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of $\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)$. After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$.

We established in Proposition 2 that the value function v_{suff}^g is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from f — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that behaves identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from f . We note that here too, it holds that a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. Hence, this preprocessing phase will give us a new function f' for which it both holds that it is equivalent to the behaviour of f for all remaining features and also satisfies that for any i it holds that both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by $\frac{1}{|D|}$ and its expected value by $\frac{1}{|D|^2}$, we obtain that $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. This implies, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|D|))$, and for the second term we derive the following:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$. This implies: $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, and therefore, $1 - |D|^2 \leq k^f \leq 0$. This yields $\frac{1}{1-k^f} \leq |D|^2$, demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly

smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

Lemma 13. *Unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble—even when $|D| = 1$. Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline $\mathbf{z} = 0_n$, deciding whether a sufficient reason exists is NP-hard unless $PTIME = NP$. Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $NP \subseteq DTIME(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ defined over a dataset $\mathbf{D}$, and some threshold δ' . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}} (f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) & \end{aligned} \quad (74)$$

where we denote $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f . Before doing so, we specify the functional form that we intend the constructed neural network f to implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, such that they agree on all coordinates outside S_j ; that is, they share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = \mathbf{y}^j \quad (76)$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — that is, no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total, this yields $2m$ vectors that form the dataset $\mathbf{D}$ from which we construct f .

We then construct a neural network f that realizes this function. The construction invokes a standard *neural network memorization* result (Vardi et al., 2022), which guarantees that a ReLU network representing any finite set of input–output pairs can be built in polynomial time. Using this result, we obtain the required f and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

We now argue that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair of vectors $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term. Moreover, because each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, there is no overlap between these pairs. Consequently, the coverage achieved by any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We first recall the definition of orthogonal DNFs (Crama & Hammer, 2011), after which the proof proceeds by adapting the decision-tree argument.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

This means that no single variable assignment can satisfy more than one term simultaneously.

The proof follows the same structural template as Theorem 1, with the essential ingredient being polynomial-time computability of v_{suff}^g for any candidate set S .

2214 For a given set S , let $\mathcal{C}^S T_i$ be the constraints that the term T_i imposes on the variables in S .
 2215 We enumerate all pairs of terms (T_i, T_j) . Any pair whose constraints on S are *inconsistent* (i.e.,
 2216 $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$) is ignored, since it contributes zero to v_{suff}^g .
 2217

2218 For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$.
 2219 The contribution of this pair to v_{suff}^g is then
 2220

$$2221 \Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

2222
 2223 Under feature independence, these probabilities factor as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Since there are at
 2224 most m^2 such term pairs, the value of v_{suff}^g can be computed in polynomial time under an independent
 2225 distribution $\mathcal{D}$. $\square$
 2226

2227 O DISCLOSURE: USAGE OF LLMs

2228
 2229 An LLM was used exclusively as a writing assistant for grammatical refinement, typo correction, and
 2230 clarity-oriented prose revision. It did not contribute to the generation of research ideas, study design,
 2231 data analysis, or interpretation of results, all of which were carried out solely by the authors.
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